

SARMAD NAWAZ KHAN

Software Engineer | AI Infrastructure & Integrations

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SUMMARY

Software engineer who grew from founding intern to engineering lead in three years at Fastn.ai, an enterprise AI-agent and integration platform. Specializes in scalable AI infrastructure: Model Context Protocol (MCP) gateways, multi-tenant enterprise architecture, and the Unified Context Layer (UCL) that lets AI agents securely reason over unified enterprise data. Ships production systems for global enterprise clients including HP, with full-stack depth across TypeScript/Node.js, React, and AWS.

EXPERIENCE

Fastn.ai

Apr 2023 – Present | Remote (US/PK)

Enterprise AI-agent and integration platform, backed by Antler and LiveOak Ventures.

Software Development Engineer II

Aug 2024 – Present

- Technical leadership:** Lead the engineering team building UCL, the product that lets AI agents securely access and reason over unified enterprise data across connected systems, owning architecture, delivery, and mentoring.
- MCP & database gateway:** Architected a standalone database gateway supporting 8+ SQL/NoSQL engines, with AI-driven schema introspection that auto-generates tools and exposes customer databases as MCP servers.
- Enterprise SDKs:** Built a headless SDK that lets enterprise customers embed the Fastn integration marketplace directly into their applications with fully custom UI/UX.
- Release management:** Designed and shipped the “Project Release” promotion pipeline, moving workflows and resources across isolated Dev/Staging/Prod environments.
- AI automation:** Created internal AI agents that automate connector generation and platform maintenance, sharply reducing manual engineering overhead.

Software Development Engineer I

May 2023 – Aug 2024

- Enterprise widget engine:** Built the embeddable widget architecture adopted by major clients (HP WXP, Perigon), enabling copy-paste integration with secure OAuth authentication flows.
- Multi-tenant architecture:** Designed a high-performance multi-tenant PostgreSQL system with strict per-project data isolation.
- Identity & security:** Engineered a custom authentication system supporting external identity providers (IdP) and granular role-based access control (RBAC).
- Platform velocity:** Optimized the core workflow engine and logging security; built a Postman Collection importer that auto-generates platform connectors, accelerating integration delivery.
- Cloud infrastructure:** Managed global frontend delivery on AWS CloudFront and S3, ensuring low-latency access worldwide.

Software Engineer Intern • Founding Team

Apr 2023 – May 2023

- Joined as one of the first three engineers; built the MVP user interface from scratch using React, TypeScript, Vite, and Apollo GraphQL.
- Established the initial CI/CD pipelines and hosting infrastructure, laying the groundwork for the platform's engineering culture.

TECHNICAL SKILLS

Languages	TypeScript, JavaScript, Python, Java, SQL, HTML/CSS
Frameworks	Node.js, NestJS, Micronaut, React, Next.js, GraphQL, FastAPI, Vite
AI & Architecture	Model Context Protocol (MCP), AI agents, RAG, workflow automation & webhooks, multi-tenancy, OAuth 2.0, RBAC
Cloud & DevOps	AWS (CloudFront, S3, ECS, Lambda, RDS), Docker, Terraform, CI/CD
Databases	PostgreSQL, MongoDB, DynamoDB, Redis

EDUCATION

University of London, BSc Computer Science

Islamabad Model College, Intermediate in Computer Science (HSSC)

London, UK

Islamabad, Pakistan